

Shuaijun Wang

ROBOTICS ALGORITHM ENGINEER · DEXTEROUS MANIPULATION

Beijing, China

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Summary

Robotics researcher and Ph.D. in robotic learning, specializing in dexterous manipulation, tactile grasping, reinforcement learning, and embodied AI. His research interests cover simulation, data collection, policy learning, sim-to-real transfer, and deployment on humanoid platforms. Research experience spans HIT, the University of Edinburgh, Tencent Robotics X, and Xiaomi Robotics.

Education

Harbin Institute of Technology / The University of Edinburgh (CSC)

PH.D. IN ROBOTIC LEARNING

Harbin / Edinburgh

Sep. 2018 – May. 2024

- Joint Ph.D. program funded by China Scholarship Council (CSC).
- Supervised by Prof. Lining Sun and Prof. Alex Li.
- Researched adaptive learning and control for robotic grasping and manipulation.
- Visiting researcher at the Edinburgh Centre for Robotics; published in *IEEE/ASME Transactions on Mechatronics*.
- Collaborated with Prof. Qiang Li on calibration-free visual servoing, published in *International Journal of Humanoid Robotics*.

Harbin Institute of Technology (Shenzhen)

M.SC. IN MECHATRONIC ENGINEERING

Shenzhen, China

Sep. 2013 – Sep. 2015

- Supervised by Prof. Xin Wang; focused on modular robotic arm design, kinematics, and manipulator control.

Northeast Forestry University (Project 211)

B.ENG. IN INTELLIGENT EQUIPMENT ENGINEERING (TOP 5%)

Harbin, China

Sep. 2009 – Sep. 2013

- Awarded Outstanding Graduate.

Work Experience

Xiaomi Robotics

SENIOR ALGORITHM ENGINEER

Beijing, China

Jul. 2024 – Present

- Selected for Xiaomi Future Star, the company's top-tier talent program.

Tencent Robotics X Lab

RESEARCH INTERN

Shenzhen, China

Dec. 2022 – Jun. 2023

- Trained PPO-based in-hand manipulation policies in MuJoCo and transferred them to real robots for tool-based stirring.
- Integrated fingertip tactile sensing and used CMA-ES for PD parameter tuning to reduce the sim-to-real gap.
- Collected visuo-tactile grasping data and explored multimodal policy learning for tactile-guided grasping.
- Tencent Robotics X PR: [video link](#).

Shenzhen Institute of Aerospace Technology Innovation

ALGORITHM ENGINEER (ROBOTICS)

Shenzhen, China

Jul. 2016 – Aug. 2018

- Developed QP-based kinematics, trajectory planning, and force-sensor-based teaching by demonstration.
- Built model-based grasping pipelines with Mask R-CNN for segmentation and 4PCS for pose estimation.
- Integrated RealSense cameras with UR arms for hand-eye calibration, perception, and robot control.
- Video: [YouTube link](#).

AVIC 613 Research Institute

ENGINEER

Luoyang, China

Sep. 2015 – Apr. 2016

- Mechanical design and fluid simulation for electro-optical radar systems.

Project Experience

Dexterous Manipulation Algorithm Development

TECHNICAL LEAD

Beijing, China

Jul. 2024 – Present

- Technical lead for dexterous manipulation, responsible for exploring core technologies including in-hand manipulation, visuo-tactile manipulation, and VLA/WAM-based embodied manipulation.

Visual Servoing for Adaptive Arm-Hand Grasping

LEAD RESEARCHER

Shenzhen, China

Mar. 2022 – Dec. 2022

- Developed calibration-free adaptive grasping using only low-cost RGB cameras, without hand-eye or camera calibration.
- Designed an end-to-end visual servoing framework for real-time 3D grasping across diverse objects.
- Published in *International Journal of Humanoid Robotics*.
- Video: Bilibili link.

Reinforcement Learning for Contact-Aware Object Manipulation

LEAD RESEARCHER

Edinburgh / Shenzhen

Dec. 2021 – Jul. 2022

- Developed a Soft Actor-Critic policy for contact-aware robotic pushing under different observation and reward settings.
- Improved generalization and training efficiency through explicit contact modeling.
- Published in *Frontiers in Neurorobotics*.
- Video: Bilibili link.

Adaptive Dexterous Hand Grasp Control via Learning from Human Demonstration

LEAD RESEARCHER

Edinburgh, UK

Sep. 2019 – Dec. 2020

- Collected human teleoperation demonstrations and trained supervised models for adaptive dexterous grasping.
- Evaluated neural architectures for object variability in size, geometry, and stiffness.
- Published in *IEEE/ASME Transactions on Mechatronics*.
- Video: YouTube link.

QP-based Motion Planning for UR Robot

DEVELOPER

Shenzhen, China

Dec. 2016 – Sep. 2017

- Designed a QP-based framework for Cartesian and joint-space trajectory tracking.
- Integrated joint-limit handling and obstacle avoidance; demonstrated tracking by writing the Chinese character Gong.
- Published in *ICARM 2017*.

Publications & Patents

1. **Wang, Shuaijun** (Corresponding author), et al. “*TacRefineNet: Tactile-Only Grasp Refinement Between Arbitrary In-Hand Object Poses.*” arXiv:2509.25746, 2025. (IEEE RAL, under review)
2. Zhou, Haoran, Yangwei You, and **Wang, Shuaijun** (Corresponding author). “*DyDexHandover: Human-like Bimanual Dynamic Dexterous Handover using RGB-only Perception.*” arXiv:2509.17350, 2025.
3. **Wang, Shuaijun**, Lining Sun*, et al. “*Learning Adaptive Grasping From Human Demonstrations.*” **IEEE/ASME Transactions on Mechatronics (TMECH)**, 2022. (SCI Q1)
4. **Wang, Shuaijun**, Lining Sun, et al. “*Seeing is Grasping: A Novel Arm-Gripper Grasping Synergy Framework without any Calibration.*” **International Journal of Humanoid Robotics (IJHR)**, 2024. (SCI)
5. **Wang, Shuaijun**, Lining Sun, Fusheng Zha, Wei Guo, Pengfei Wang. “*Learning Adaptive Reaching and Pushing Skills Using Contact Information.*” **Frontiers in Neurorobotics**, 2023. (SCI Q2)
6. **Wang, Shuaijun**, et al. “*A Generic Control Method of Manipulator Based on Optimization.*” In **Proc. of the 2nd IEEE ICARM**, 2017. (EI Indexed)
7. **Wang, Shuaijun**, Zhang, B., Wang, X., et al. “*Design and Research of a Novel Fast Connecting and Separating Modular Space Manipulator.*” **Applied Mechanics & Materials**, 2015. (EI Indexed)
8. Wu, Jingyuan, Xin Wang, **Wang, Shuaijun**, Shuaishuai Wang, Bo Zhan, Fei Chen. “*An Optimization-Based Direct Teaching Method for Manipulator with Tool’s Gravity Compensation.*” In **IEEE CYBER**, 2019. (EI Indexed)
9. Wang, Xueqian; **Wang, Shuaijun**; Zhang, Bo; Liang, Bin; Zou, Yu. “*Modular Space Robotic Arm Joints and Robotic Arms That Can Be Used for Rapid Reconfiguration.*” **Chinese Patent**, CN104647396A, 2015.

Academic Services

Conferences: ICRA (2024), IROS (2021, 2022), ICARM (2019)

REVIEWER

Journals: IEEE RAL; IEEE/ASME TMECH; International Journal of Humanoid Robotics; Industrial Robot; Robotic Intelligence and Automation

REVIEWER

Invited Talks

Beijing Academy of Artificial Intelligence · Embodied Intelligence Research Center Beijing, China
ADAPTIVE ROBOTIC GRASPING AND MANIPULATION VIA EMBODIED INTELLIGENCE Sep. 2023

- Invited by Prof. He Wang.

Shenzhen Technology University · School of Big Data and Internet Shenzhen, China
EMBODIED INTELLIGENCE METHODS FOR GENERAL-PURPOSE ROBOTIC GRASPING AND MANIPULATION Dec. 2023

- Invited by Prof. Qiang Li; received Assistant Professor offer following the talk.

Honors & Awards

2011 National Encouragement Scholarship, Ministry of Education, P.R. China China
2013 Outstanding Graduate, Northeast Forestry University Harbin, China
2019–2021 CSC Government Scholarship for Overseas Study, China Scholarship Council (CSC) China